

# Chapter 1-introduction to dynamics

- Dynamics
  - Is a branch of mechanics which deals with motion of bodies that are subjected to some force
  - It has got two distinct parts
    - Kinematics
    - Kinetics
- Kinematics
  - Study of motion without reference to the force which cause the motion
- Kinetics
  - Is the study of motion with reference to the force which is the cause of the motion

- There are basic concepts that should be known before further discussion
  1. Space –is the geometric region occupied by bodies
    - Rectangular coordinate system( $x,y,z$ )
    - Polar coordinate system( $r,$
    - Normal-tangent
  2. Particle –is a body of negligible dimension. When the dimension of the body is irrelevant to the description of its motion or the action of force on it.
  3. Rigid body-is a body whose changes are negligible compared with the overall dimension of the body or with changes in position of the body as a whole.

# Newton's law

- **1<sup>ST</sup> law**- a particle remains at rest or continues to move in a straight line with constant velocity if there is no resultant-force acting on it.
- **2<sup>nd</sup> law** – a particle acted upon by a resultant force moves in such manner that the time rate of change of its linear momentum is equal to the force
- $\sum F = ma$  for a single particle  
where F is the force, m is the mass, and a is the acceleration
- **3<sup>rd</sup> law**-forces that result from interactions of particles and such forces between two particles are equal in magnitude, opposite in direction, and collinear

# Inertial frame of reference

- **Newton II**:-requires the concept of an “internal frame of reference” because it depends on the reference frame in which motion is observed (this is not so far mass of particle or the force applied to the particles)
  - An internal reference frame must be non-accelerating .
  - What is non accelerating ? (newton I)
- **Absolute inertial frame**-It is where newton I holds and is idealized situation. Surface of earth is good enough(for short periods of time-minutes).Earth is rotating if particles followed long enough, will curve

# Engineering Mechanics: Dynamics

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- Dynamics
  - Branch of mechanics that deals with the motion of bodies under the action of forces  
***(Accelerated Motion)***
- Two distinct parts:
  - Kinematics
    - study of motion without reference to the forces that cause motion or are generated as a result of motion
  - Kinetics
    - relates the action of forces on bodies to their resulting motions

# Engineering Mechanics: Dynamics

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- Basis of rigid body dynamics
  - Newton's 2<sup>nd</sup> law of motion
    - A particle of mass “ $m$ ” acted upon by an unbalanced force “ $F$ ” experiences an acceleration “ $a$ ” that has the **same direction as the force** and a **magnitude that is directly proportional to the force**
    - $a$  is the resulting **acceleration** measured in a **non-accelerating frame of reference**

# Engineering Mechanics: Dynamics

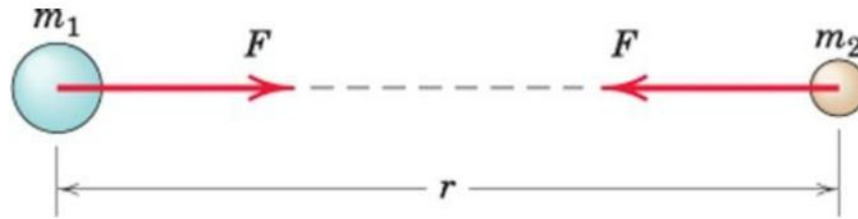
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- Space
  - Geometric region occupied by bodies
    - *Reference system*
      - Linear or angular measurements
    - **Primary reference system or astronomical frame of reference**
      - Imaginary set of rectangular axes fixed in space
      - Validity for measurements for velocity  $<$  speed of light
      - Absolute measurements
    - **Reference frame attached to the earth??**
- Time
- Mass

# Engineering Mechanics: Dynamics

- **Newton's law of gravitation**

$$F = G \frac{m_1 m_2}{r^2}$$



- **$F$**  :: mutual force of attraction between two particles
- **$G$**  :: universal gravitational constant
  - $6.673 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$

# Engineering Mechanics: Dynamics

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- **Weight**

- Only significant gravitational force between the earth and a particle located near the surface

$$W = G \frac{mM_e}{r^2}$$

$$W = mg$$

- $g = GM_e/r^2 ::$  acceleration due to gravity (9.81m/s<sup>2</sup>)
- Variation of  $g$  with altitude

$$g = g_0 \frac{R^2}{(R + h)^2}$$

$g$  is the absolute acceleration due to gravity at altitude  $h$

$g_0$  is the absolute acceleration due to gravity at sea level

$R$  is the radius of the earth

# Engineering Mechanics: Dynamics

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## Effect of Altitude on Gravitation

- **Force of gravitational attraction of the earth on a body depends on the position of the body relative to the earth**
- **Assuming the earth to be a perfect homogeneous sphere, a mass of 1 kg would be attracted to the earth by a force of:**
  - **9.825 N** if the mass is on the surface of the earth
  - **9.822 N** if the mass is at an altitude of 1 km
  - **9.523 N** if the mass is at an altitude of 100 km
  - **7.340 N** if the mass is at an altitude of 1000 km
  - **2.456 N** if the mass is at an altitude of equal to the mean radius of the earth, 6371 km

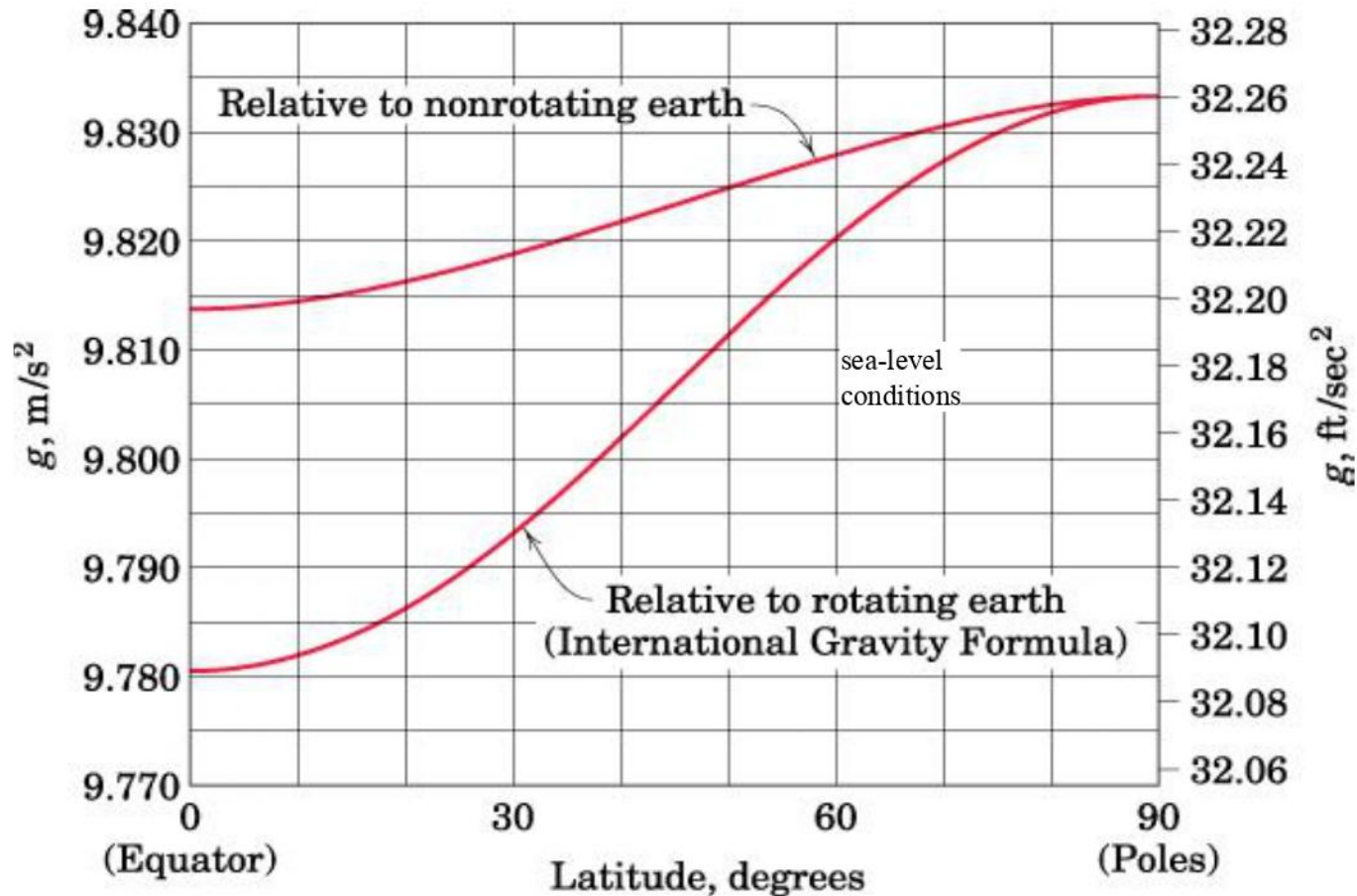
# Engineering Mechanics: Dynamics

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- Effect of earth's rotation
  - $g$  from law of gravitation
    - Fixed set of axes at the centre of the earth
      - Absolute value of  $g$
    - Earth's rotation
      - **Actual acceleration of a freely falling body is less than absolute  $g$**
      - **Measured from a position attached to the surface of the earth**

# Engineering Mechanics: Dynamics

- Effect of earth's rotation



– Engineering applications :: variation of  $g$  is ignored

# Kinematics of Particles

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- **Motion**

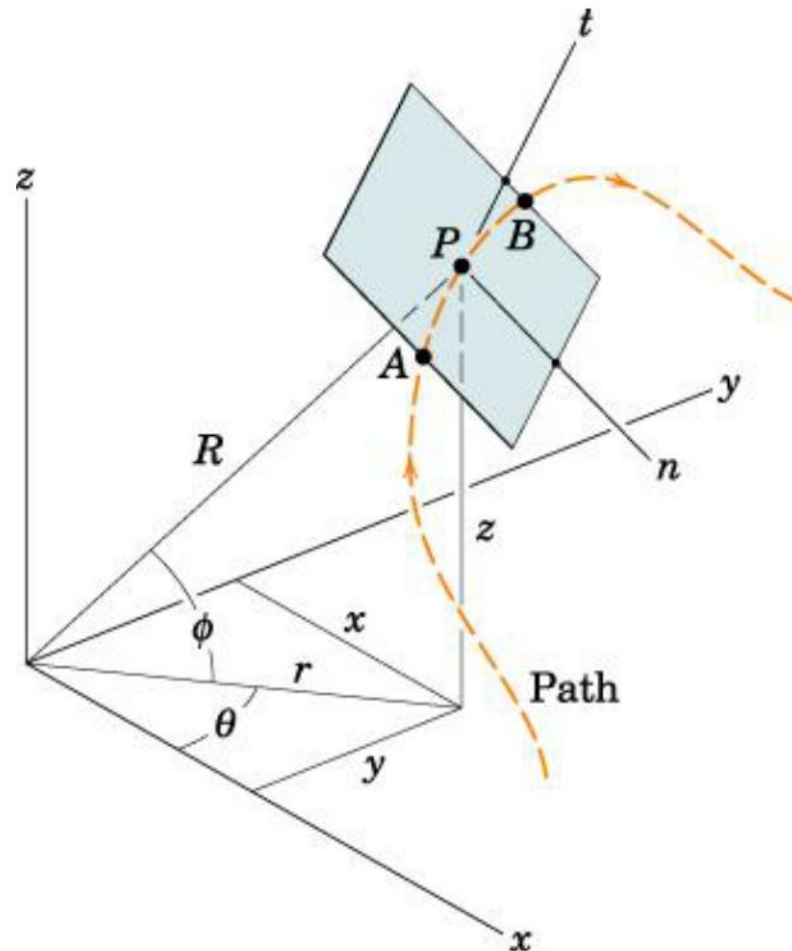
- Constrained :: confined to a specific path
- Unconstrained :: not confined to a specific path

- **Choice of coordinates**

- Position of P at any time  $t$ 
  - rectangular (i.e., Cartesian) coordinates  $x, y, z$
  - cylindrical coordinates  $r, \theta, z$
  - spherical coordinates  $R, \theta, \phi$
- Path variables
  - Measurements along the tangent  $t$  and normal  $n$  to the curve

# Kinematics of Particles

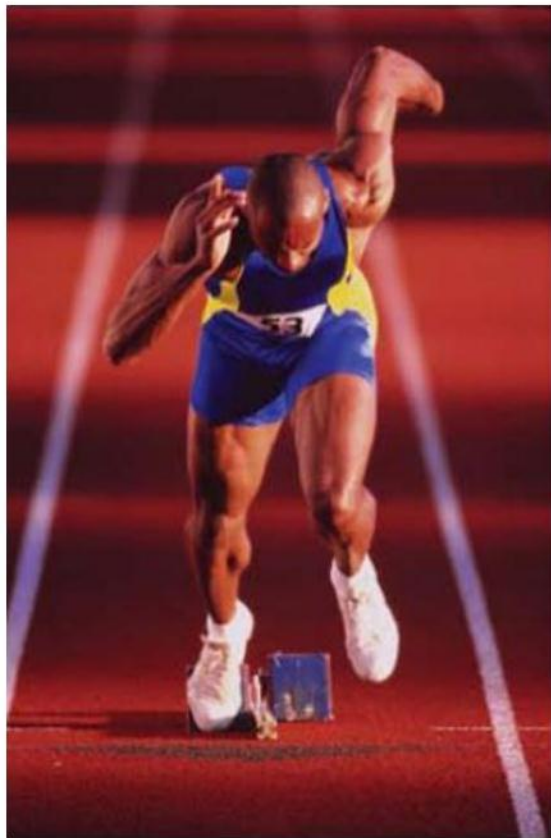
- Choice of coordinates



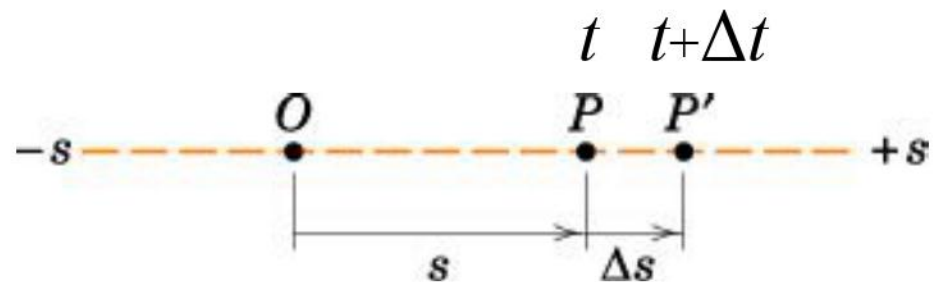
# Kinematics of Particles

## Rectilinear Motion

- Motion along a straight line

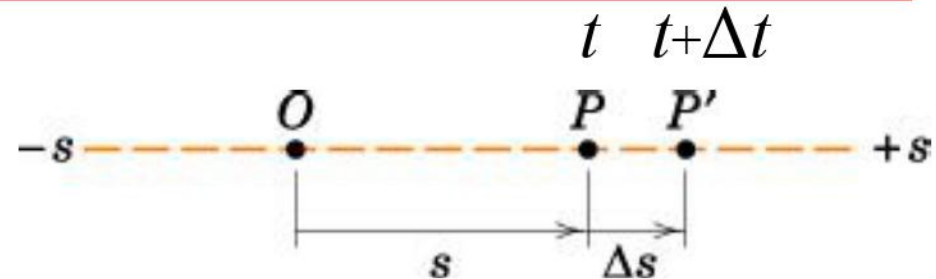


**This sprinter will undergo rectilinear acceleration until he reaches his terminal speed.**



# Kinematics of Particles :: Rectilinear Motion

- Motion along a straight line



Position at any instance of time  $t$

:: specified by its distance  $s$  measured from some convenient reference point  $O$  fixed on the line

:: (disp. is negative if the particle moves in the negative  $s$ -direction).

Velocity of the particle:

$$v = \frac{ds}{dt} = \dot{s}$$

Both are  
vector quantities

+ve or -ve depending  
on +ve or -ve displacement

Acceleration of the particle:

$$a = \frac{dv}{dt} = \dot{v} \quad \text{or} \quad a = \frac{d^2s}{dt^2} = \ddot{s}$$

+ve or -ve depending on whether  
velocity is increasing or decreasing

$$v dv = a ds \quad \text{or} \quad \dot{s} d\dot{s} = \ddot{s} ds$$

# Kinematics of Particles

## Rectilinear Motion: Graphical Interpretations

Using **s-t** curve, **v-t** & **a-t** curves can be plotted.

Area under **v-t** curve during time  $dt = vdt == ds$

- Net disp from  $t_1$  to  $t_2$  = corresponding area under v-t curve  $\rightarrow$

$$\int_{s_1}^{s_2} ds = \int_{t_1}^{t_2} vdt$$

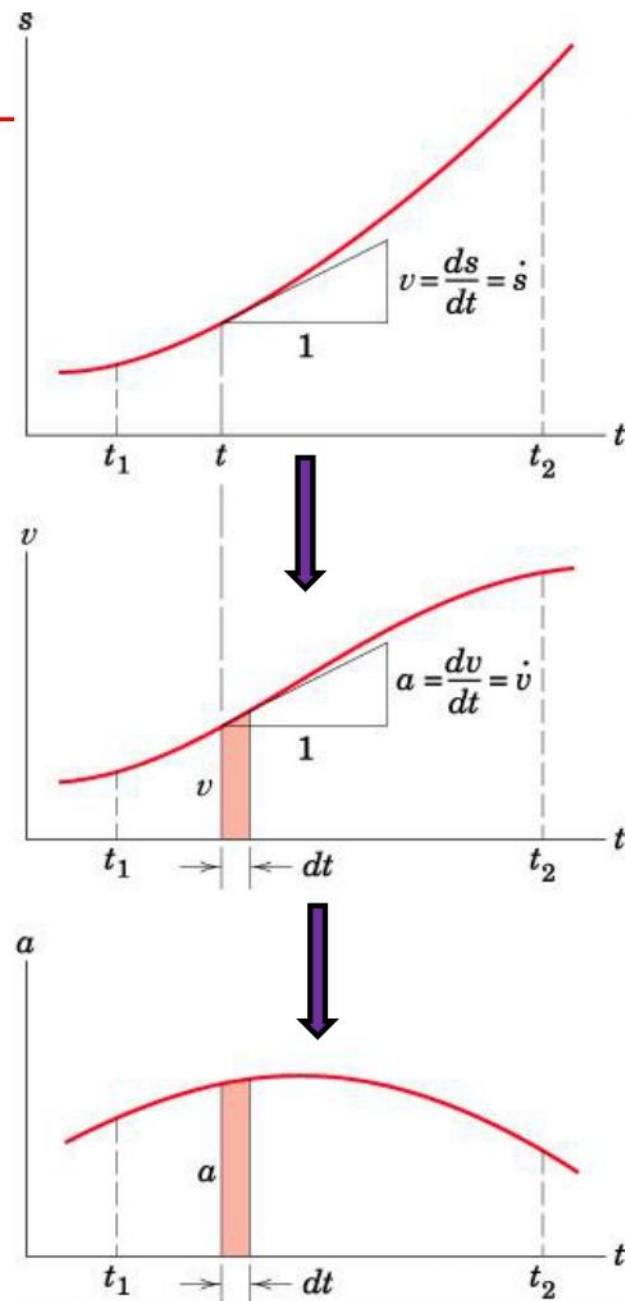
or  $s_2 - s_1 = (\text{area under v-t curve})$

Area under **a-t** curve during time  $dt = adt == dv$

- Net change in vel from  $t_1$  to  $t_2$  = corresponding area under a-t curve  $\rightarrow$

$$\int_{v_1}^{v_2} dv = \int_{t_1}^{t_2} adt$$

or  $v_2 - v_1 = (\text{area under a-t curve})$



# Kinematics of Particles

## Rectilinear Motion: Graphical Interpretations

Two additional graphical relations:

Area under  $a$ - $s$  curve during disp  $ds = \int a ds = \int v dv$

- Net area under  $a$ - $s$  curve betn position coordinates  $s_1$  and  $s_2 \rightarrow$

$$\int_{v_1}^{v_2} v dv = \int_{s_1}^{s_2} a ds$$

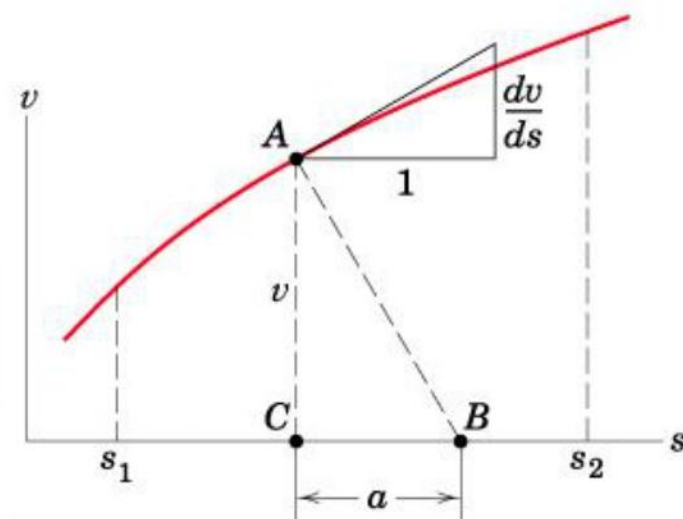
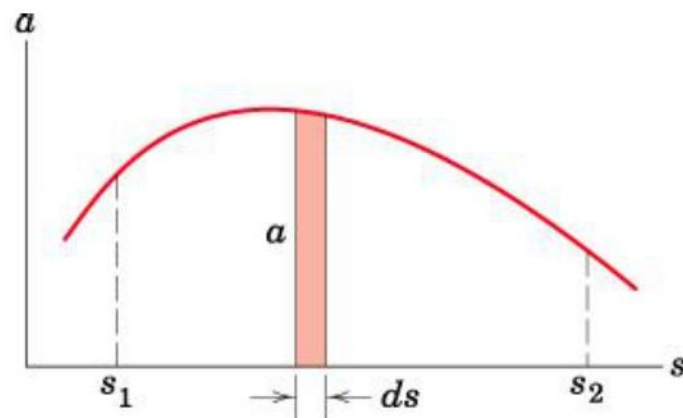
$$\text{or } \frac{1}{2} (v_2^2 - v_1^2) = (\text{area under } a\text{-}s \text{ curve})$$

Slope of  $v$ - $s$  curve at any point  $A = dv/ds$

- Construct a normal  $AB$  to the curve at  $A$ . From similar triangles:

$$\frac{\overline{CB}}{v} = \frac{dv}{ds} \rightarrow \overline{CB} = v \frac{dv}{ds} = a \quad (\text{acceleration})$$

- Vel and posn coordinate axes should have the same numerical scales so that the accn read on the x-axis in meters will represent the actual accn in  $\text{m/s}^2$



# Kinematics of Particles :: Rectilinear Motion

## Analytical Integration to find the position coordinate

Acceleration may be specified as a function of time, velocity, or position coordinate, or as a combined function of these.

### (a) Constant Acceleration

At the beginning of the interval  $\rightarrow t = 0, s = s_0, v = v_0$

For a time interval  $t$ : integrating the following two equations

$$a = \frac{dv}{dt}$$

$$v dv = a ds$$

$$\int_{v_0}^v dv = a \int_0^t dt \quad \text{or} \quad v = v_0 + at$$
$$\int_{v_0}^v v dv = a \int_{s_0}^s ds \quad \text{or} \quad v^2 = v_0^2 + 2a(s - s_0)$$

Substituting in the following equation and integrating will give the position coordinate:

$$v = \frac{ds}{dt}$$

$$\int_{s_0}^s ds = \int_0^t (v_0 + at) dt \quad \text{or} \quad s = s_0 + v_0 t + \frac{1}{2} at^2$$

Equations applicable for **Constant Acceleration** and for **time interval 0 to  $t$**

# Kinematics of Particles :: Rectilinear Motion

## Analytical Integration to find the position coordinate

(b) Acceleration given as a function of time,  $a = f(t)$

At the beginning of the interval  $\rightarrow t = 0, s = s_0, v = v_0$

For a time interval  $t$ : integrating the following equation

$$a = \frac{dv}{dt} \rightarrow f(t) = \frac{dv}{dt} \quad \int_{v_0}^v dv = \int_0^t f(t) dt \quad \text{or} \quad v = v_0 + \int_0^t f(t) dt$$

Substituting in the following equation and integrating will give the position coordinate:

$$v = \frac{ds}{dt} \quad \int_{s_0}^s ds = \int_0^t v dt \quad \text{or} \quad s = s_0 + \int_0^t v dt$$

Alternatively, following second order differential equation may be solved to get the position coordinate:

$$a = \frac{d^2s}{dt^2} = \ddot{s} \rightarrow \ddot{s} = f(t)$$

# Kinematics of Particles :: Rectilinear Motion

## Analytical Integration to find the position coordinate

(c) Acceleration given as a function of velocity,  $a = f(v)$

At the beginning of the interval  $\rightarrow t = 0, s = s_0, v = v_0$

For a time interval  $t$ : Substituting  $a$  and integrating the following equation

$$a = \frac{dv}{dt} \rightarrow f(v) = \frac{dv}{dt} \quad t = \int_0^t dt = \int_{v_0}^v \frac{dv}{f(v)}$$

Solve for  $v$  as a function of  $t$  and integrate the following equation to get the position coordinate:

$$v = \frac{ds}{dt}$$

Alternatively, substitute  $a = f(v)$  in the following equation and integrate to get the position coordinate :

$$v dv = a ds \quad \int_{v_0}^v \frac{v dv}{f(v)} = \int_{s_0}^s ds \quad \text{or} \quad s = s_0 + \int_{v_0}^v \frac{v dv}{f(v)}$$

# Kinematics of Particles: Rectilinear Motion

## Analytical Integration to find the position coordinate

(d) Acceleration given as a function of displacement,  $a = f(s)$

At the beginning of the interval  $\rightarrow t = 0, s = s_0, v = v_0$

For a time interval  $t$ : substituting  $a$  and integrating the following equation

$$v dv = a ds \quad \int_{v_0}^v v dv = \int_{s_0}^s f(s) ds \quad \text{or} \quad v^2 = v_0^2 + 2 \int_{s_0}^s f(s) ds$$

Solve for  $v$  as a function of  $s$  :  $v = g(s)$ , substitute in the following equation and integrate to get the position coordinate:

$$v = \frac{ds}{dt} \quad \int_{s_0}^s \frac{ds}{g(s)} = \int_0^t dt \quad \text{or} \quad t = \int_{s_0}^s \frac{ds}{g(s)}$$

It gives  $t$  as a function of  $s$ . Rearrange to obtain  $s$  as a function of  $t$  to get the position coordinate.

In all these cases, if integration is difficult, graphical, analytical, or computer methods can be utilized.

# Kinematics of Particles: Rectilinear Motion

## Example

Position coordinate of a particle confined to move along a straight line is given by  $s = 2t^3 - 24t + 6$ , where  $s$  is measured in meters from a convenient origin and  $t$  is in seconds. Determine: (a) time reqd for the particle to reach a velocity of 72 m/s from its initial condition at  $t = 0$ , (b) acceleration of the particle when  $v = 30$  m/s, and (c) net disp of the particle during the interval from  $t = 1$  s to  $t = 4$  s.

## Solution

$$\begin{aligned}\text{Differentiating } s = 2t^3 - 24t + 6 &\rightarrow v = 6t^2 - 24 \text{ m/s} \\ &\rightarrow a = 12t \text{ m/s}^2\end{aligned}$$

$$\begin{aligned}\text{(a) } v = 72 \text{ m/s} &\rightarrow t = \pm 4 \text{ s} \\ &\text{(- 4 s happened before initiation of motion } \rightarrow \text{ no physical interest.)} \\ &\rightarrow t = 4 \text{ s}\end{aligned}$$

$$\text{(b) } v = 30 \text{ m/s} \rightarrow t = 3 \text{ sec} \rightarrow a = 36 \text{ m/s}^2$$

$$\begin{aligned}\text{(c) } t = 1 \text{ s to } 4 \text{ s. Using } s = 2t^3 - 24t + 6 \\ \Delta s = s_4 - s_1 = [2(4^3) - 24(4) + 6] - [2(1^3) - 24(1) + 6] \\ \Delta s = 54 \text{ m}\end{aligned}$$

# Kinematics of Particles

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## Plane Curvilinear Motion

Motion of a particle along a curved path which lies in a single plane.



For a short time during take-off and landing, planes generally follow plane curvilinear motion

# Kinematics of Particles

## Plane Curvilinear Motion:

Between A and A':

Average velocity of the particle :  $\mathbf{v}_{av} = \Delta \mathbf{r} / \Delta t$

→ A vector whose direction is that of  $\Delta \mathbf{r}$  and whose magnitude is magnitude of  $\Delta \mathbf{r} / \Delta t$

Average speed of the particle =  $\Delta s / \Delta t$

*Instantaneous velocity of the particle is defined as the limiting value of the average velocity as the time interval approaches zero →*

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t}$$

→  $\mathbf{v}$  is always a vector tangent to the path

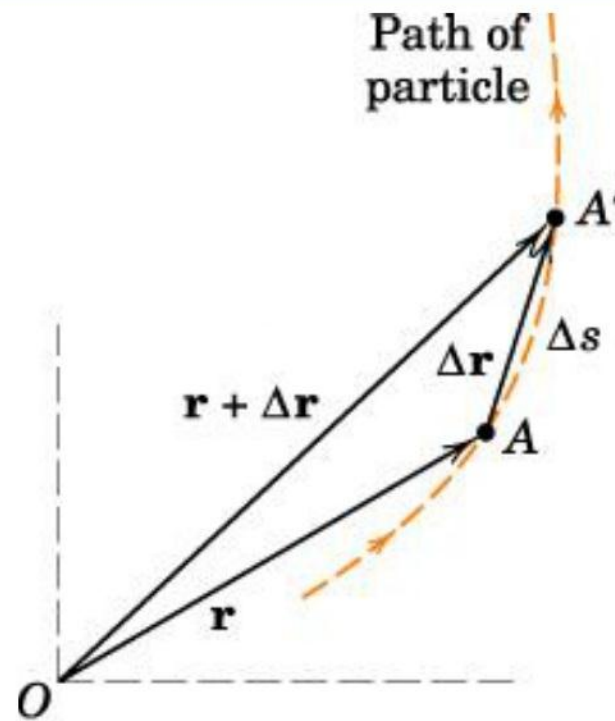
Extending the definition of derivative of a scalar to include vector quantity:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$$

Derivative of a vector is a vector having a magnitude and a direction.

Magnitude of  $\mathbf{v}$  is equal to speed (scalar)

$$v = |\mathbf{v}| = \frac{ds}{dt} = \dot{s}$$



# Kinematics of Particles

## Plane Curvilinear Motion

Magnitude of the derivative:

$$|d\mathbf{r} / dt| = |\dot{\mathbf{r}}| = \dot{s} = |\mathbf{v}| = v$$

→ Magnitude of the velocity or the speed

Derivative of the magnitude:

$$d|\mathbf{r}| / dt = dr / dt = \dot{r}$$

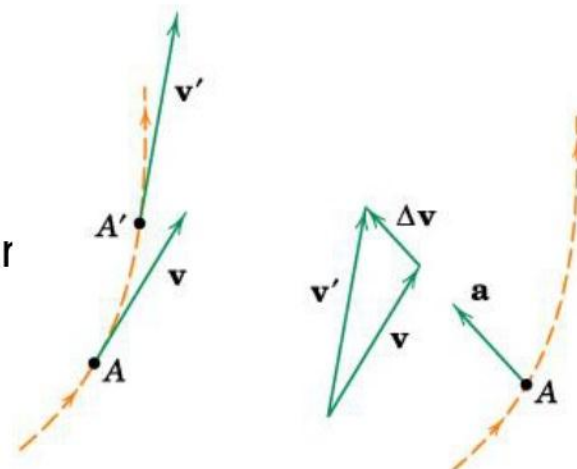
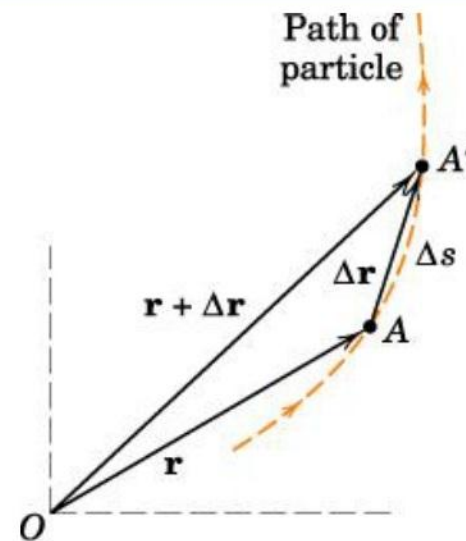
→ Rate at which the length of the position vector is changing

Velocity of the particle at A → tangent vector  $\mathbf{v}$

Velocity of the particle at A' → tangent vector  $\mathbf{v}'$

→  $\mathbf{v}' - \mathbf{v} = \Delta \mathbf{v}$

→  $\Delta \mathbf{v}$  Depends on both the change in magnitude of  $\mathbf{v}$  and on the change in direction of  $\mathbf{v}$ .



# Kinematics of Particles

## Plane Curvilinear Motion

Between A and A':

Average acceleration of the particle :  $\mathbf{a}_{av} = \Delta \mathbf{v} / \Delta t$

→ A vector whose direction is that of  $\Delta \mathbf{v}$  and whose magnitude is the magnitude of  $\Delta \mathbf{v} / \Delta t$

*Instantaneous accln of the particle is defined as the limiting value of the average accln as the time interval approaches zero →*

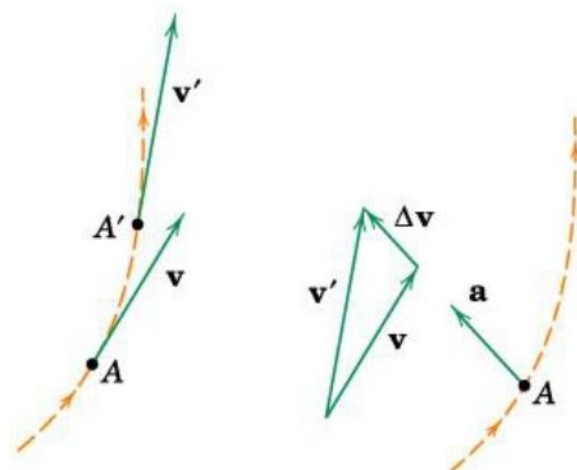
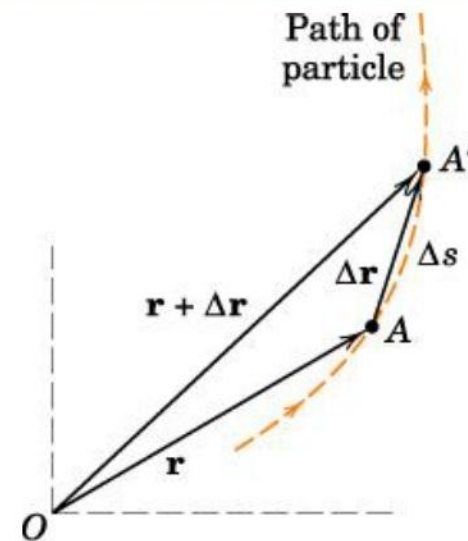
$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t}$$

By definition of the derivative:

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \dot{\mathbf{v}}$$

→ In general, direction of the acceleration of a particle in curvilinear motion neither tangent to the path nor normal to the path.

→ Acceleration component normal to the path points toward the center of curvature of the path.



There are three kinds of coordinates  
that are used to analyze plane  
curvilinear motions

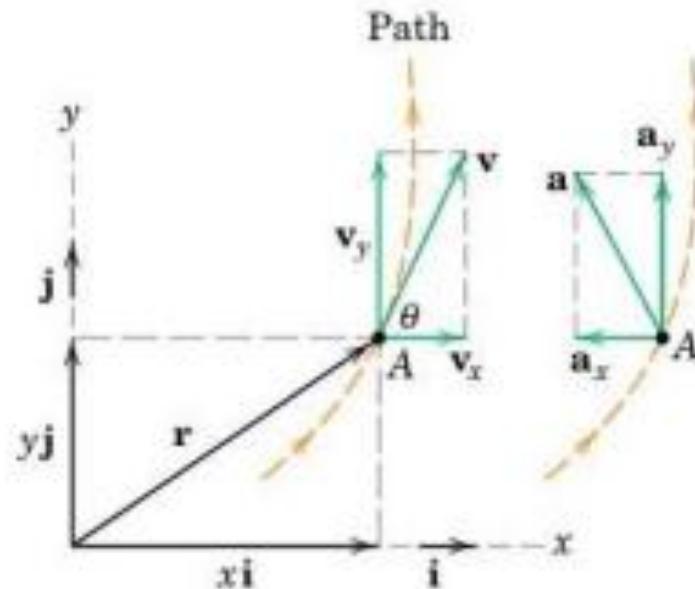
# 1. Rectangular coordinate system(x,y)

- This type of coordinate system is fixed

$$\begin{aligned}\mathbf{r} &= x\mathbf{i} + y\mathbf{j} \\ \mathbf{v} &= \dot{\mathbf{r}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} \\ \mathbf{a} &= \dot{\mathbf{v}} = \ddot{\mathbf{r}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}\end{aligned}$$

$$v_x = \dot{x}, v_y = \dot{y} \text{ and}$$

$$a_x = \dot{v}_x = \ddot{x}, a_y = \dot{v}_y = \ddot{y}.$$

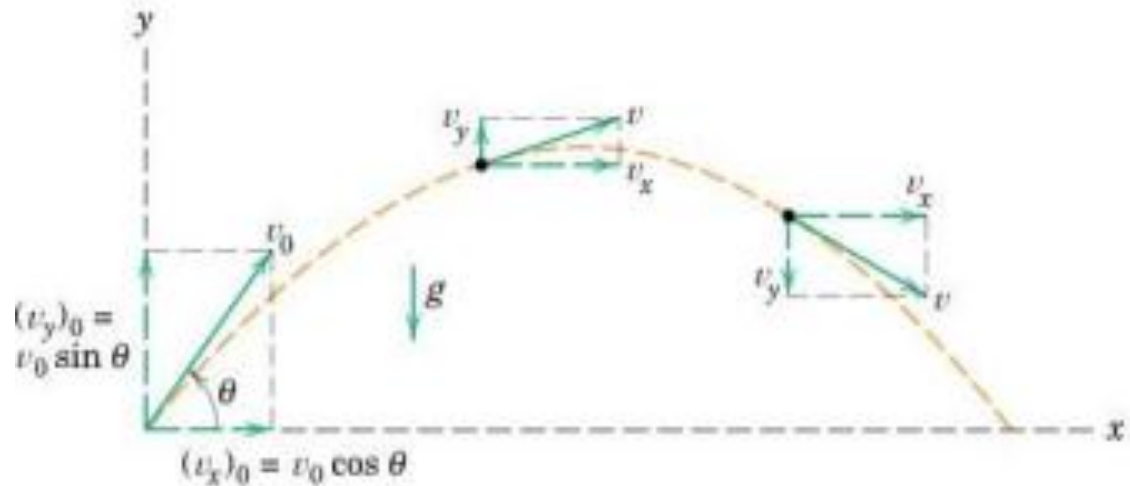


$$v^2 = v_x^2 + v_y^2 \quad v = \sqrt{v_x^2 + v_y^2} \quad \tan \theta = \frac{v_y}{v_x}$$

$$a^2 = a_x^2 + a_y^2 \quad a = \sqrt{a_x^2 + a_y^2}$$

# Projectile motion

- The type of motion
  - $a_x=0$
  - $a_y=-g$
  - $g=9.81\text{m/s}^2$



$$v_x = (v_x)_0$$

$$v_y = (v_y)_0 - gt$$

$$x = x_0 + (v_x)_0 t$$

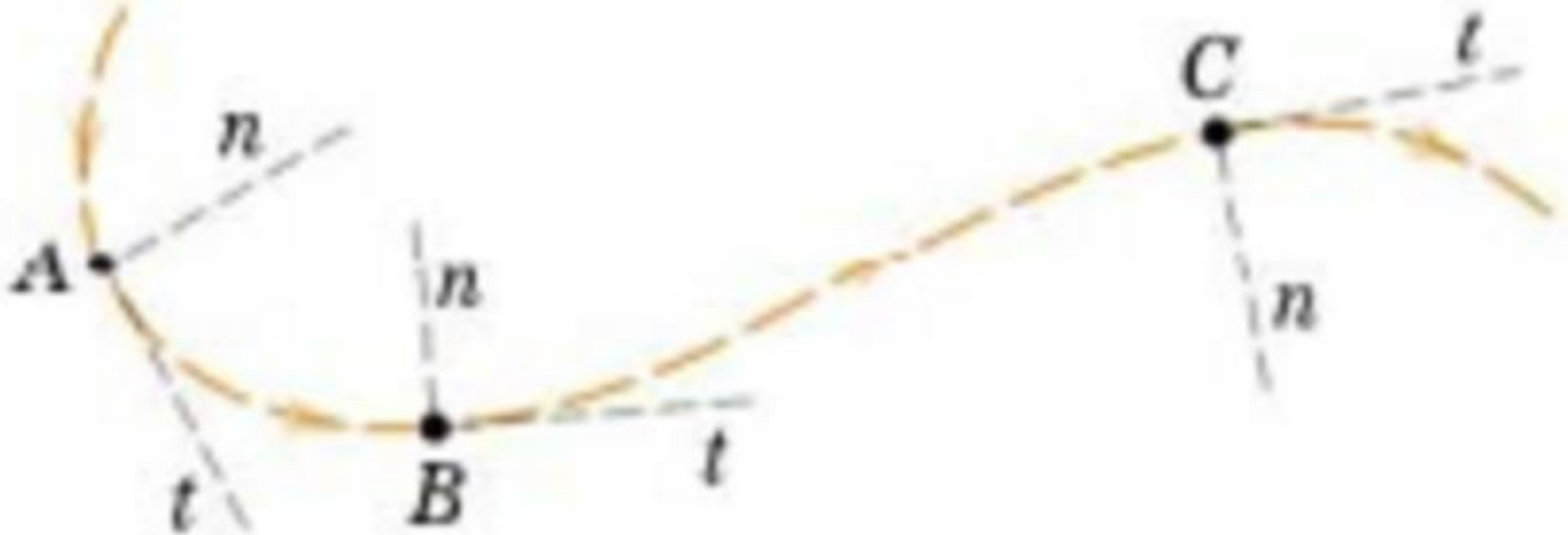
$$y = y_0 + (v_y)_0 t - \frac{1}{2}gt^2$$

$$v_y^2 = (v_y)_0^2 - 2g(y - y_0)$$

# Week 3

## 2. Normal -tangential coordinates( $n,t$ )

- movable type of coordinate system



- Velocity and acceleration in n-t

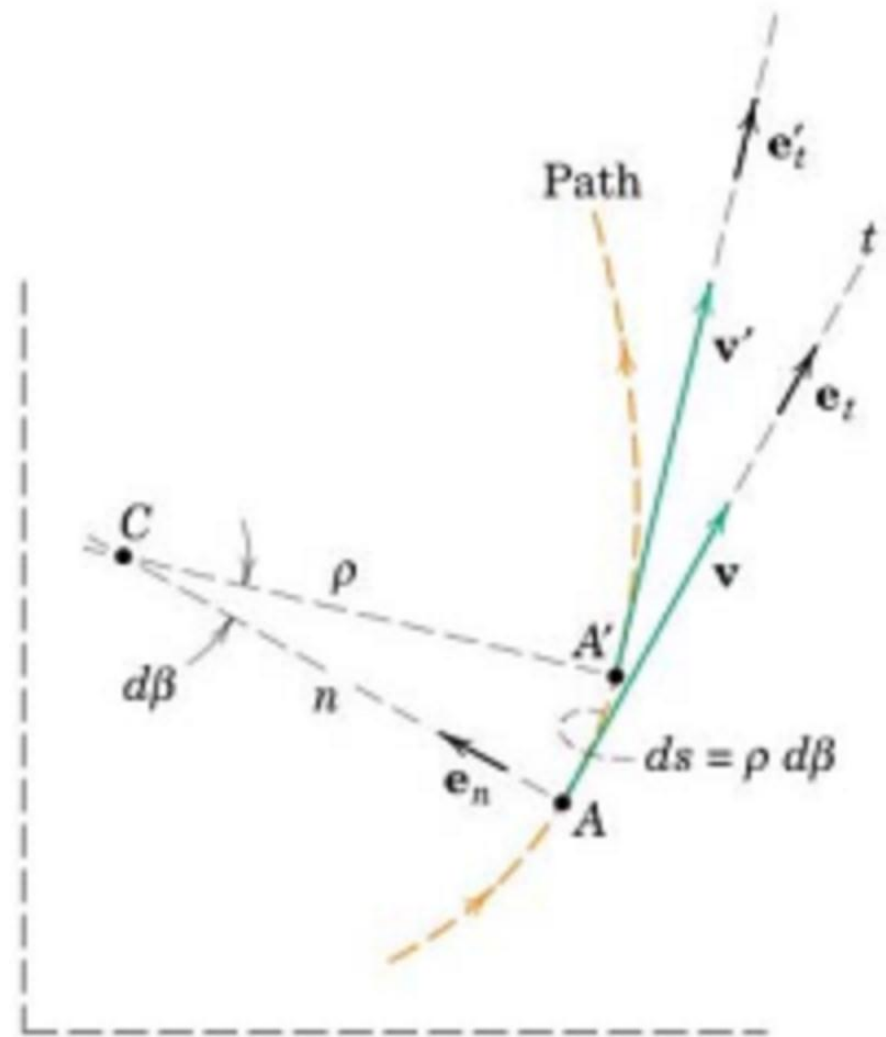
$$ds = \rho d\beta,$$

$$v = ds/dt = \rho d\beta/dt,$$

$$\mathbf{v} = \rho \dot{\beta} \mathbf{e}_t \quad \text{-----} (1)$$

$$\mathbf{v} = v \mathbf{e}_t \quad \text{-----} (2)$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d(v\mathbf{e}_t)}{dt} = v\dot{\mathbf{e}}_t + \dot{v}\mathbf{e}_t \quad \text{-----} (3)$$



(a)

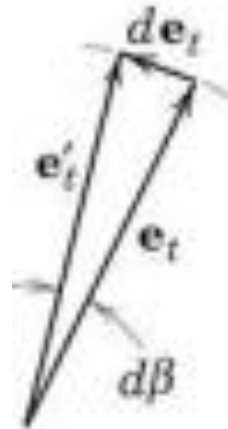
- Unit vector derivation to substitute acceleration component

$$d\mathbf{e}_t = |\mathbf{e}_t| d\beta = d\beta$$

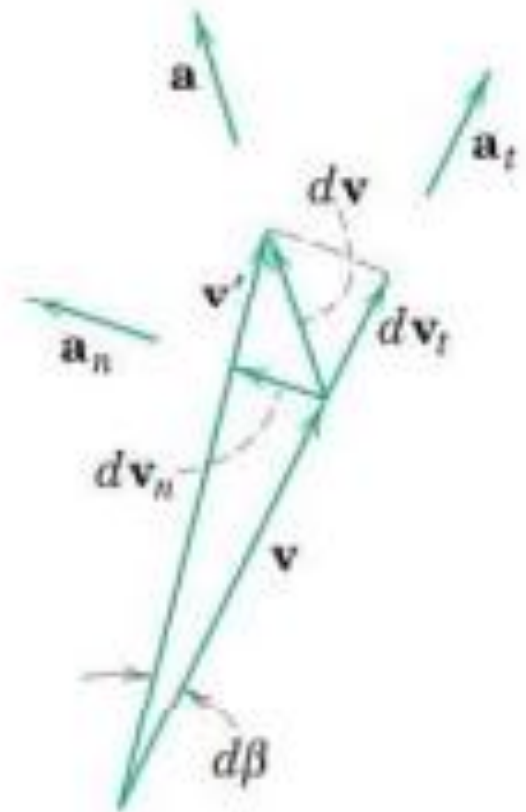
$$d\mathbf{e}_t = \mathbf{e}_n d\beta. \quad \frac{d\mathbf{e}_t}{d\beta} = \mathbf{e}_n$$

$$d\mathbf{e}_t/dt = (d\beta/dt)\mathbf{e}_n,$$

$$\dot{\mathbf{e}}_t = \dot{\beta}\mathbf{e}_n \quad (4)$$



(b)



(c)

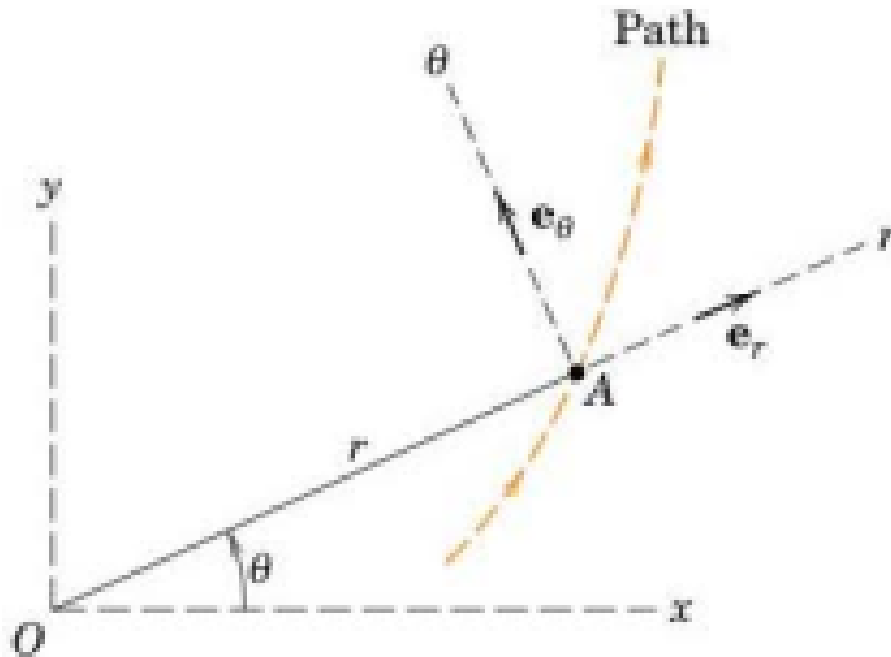
$$\mathbf{a} = \frac{v^2}{\rho} \mathbf{e}_n + \dot{v} \mathbf{e}_t \quad \text{----- final acc equation}$$

$$a_n = \frac{v^2}{\rho} = \rho \dot{\beta}^2 = v \dot{\beta}$$

$$a_t = \dot{v} = \ddot{s}$$

$$a = \sqrt{a_n^2 + a_t^2}$$

### 3. Polar coordinates( $r, \theta$ )



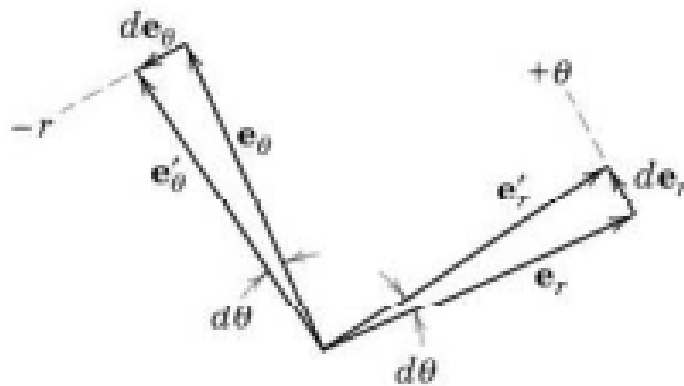
(a)

-point  $p$  can be specified by position vectors  $r$

$$\mathbf{r} = r\mathbf{e}_r$$

- To get the velocity, we will derivate  $\mathbf{r}$  with respect to time

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r$$



(b)

$$d\mathbf{e}_r = \mathbf{e}_\theta d\theta \text{ and } d\mathbf{e}_\theta = -\mathbf{e}_r d\theta.$$

$$\frac{d\mathbf{e}_r}{d\theta} = \mathbf{e}_\theta \quad \text{and} \quad \frac{d\mathbf{e}_\theta}{d\theta} = -\mathbf{e}_r$$

If, on the other hand, we divide them by  $dt$ , we have  $d\mathbf{e}_r/dt = (d\theta/dt)\mathbf{e}_\theta$  and  $d\mathbf{e}_\theta/dt = -(d\theta/dt)\mathbf{e}_r$ , or simply

$$\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta \quad \text{and} \quad \dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r \quad \text{-----} (2)$$

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r$$

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta \quad \text{-----} (2)$$

$$\begin{aligned} v_r &= \dot{r} \\ v_\theta &= r\dot{\theta} \\ v &= \sqrt{v_r^2 + v_\theta^2} \end{aligned}$$

- Acceleration:

$$\mathbf{a} = \dot{\mathbf{v}} = (\ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r) + (\dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta)$$

Substitution of  $\dot{\mathbf{e}}_r$  and  $\dot{\mathbf{e}}_\theta$  from Eq. 1 and collecting terms give

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta \text{ ----- (3)}$$

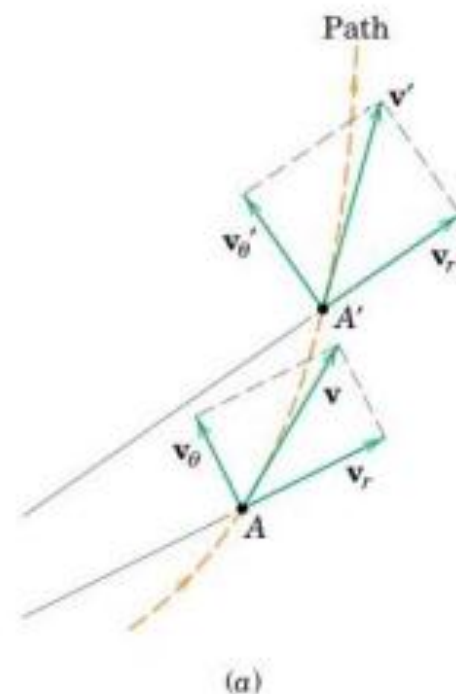
$$a_r = \ddot{r} - r\dot{\theta}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

$$a = \sqrt{a_r^2 + a_\theta^2}$$

We can write the  $\theta$ -component alternatively as

$$a_\theta = \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$$



## 4. Relative motions(translating axes)

- Vector representation

$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}_{A/B}$$

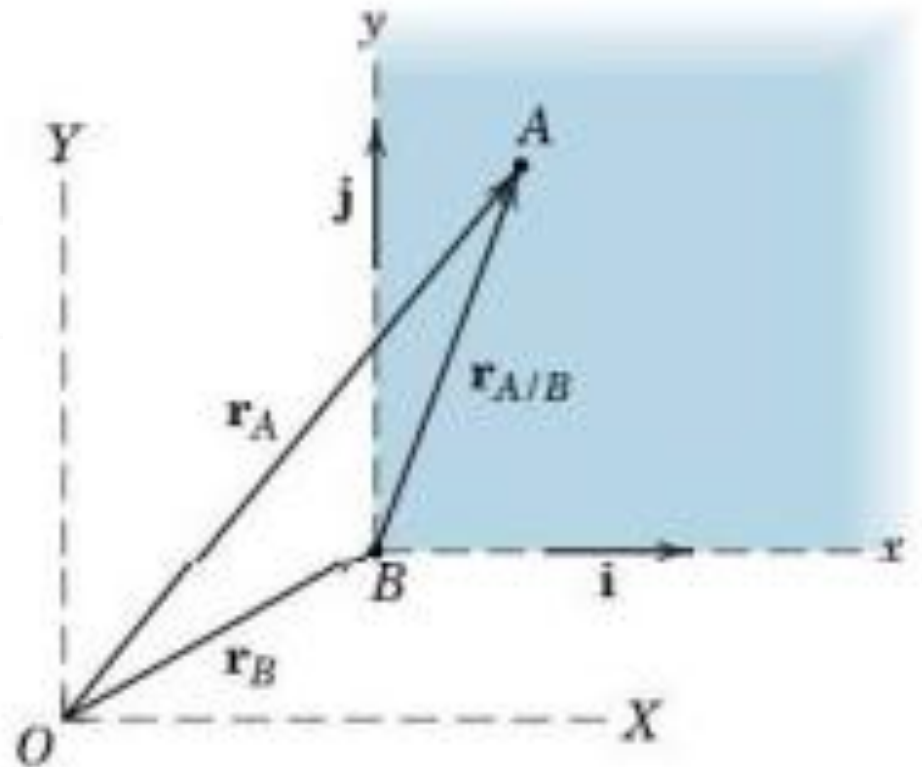
$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathbf{r}}_{A/B} \quad \text{or} \quad \boxed{\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}} \quad 1$$

$$\ddot{\mathbf{r}}_A = \ddot{\mathbf{r}}_B + \ddot{\mathbf{r}}_{A/B} \quad \text{or} \quad \boxed{\mathbf{a}_A = \mathbf{a}_B + \mathbf{a}_{A/B}} \quad 2$$

$$\dot{\mathbf{r}}_{A/B} = \mathbf{v}_{A/B} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}.$$

- Taking the second derivative with respect to time

$$\ddot{\mathbf{r}}_{A/B} = \dot{\mathbf{v}}_{A/B} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}.$$

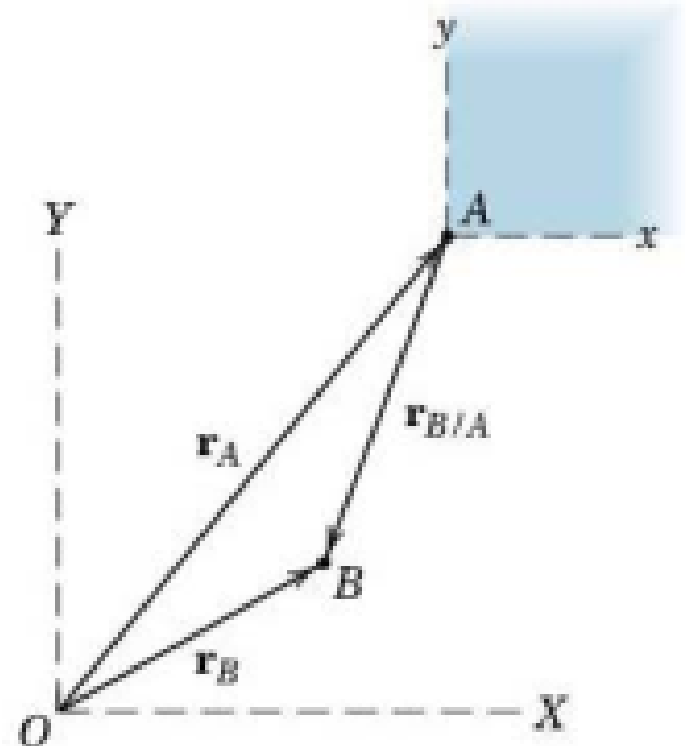


## Additional considerations

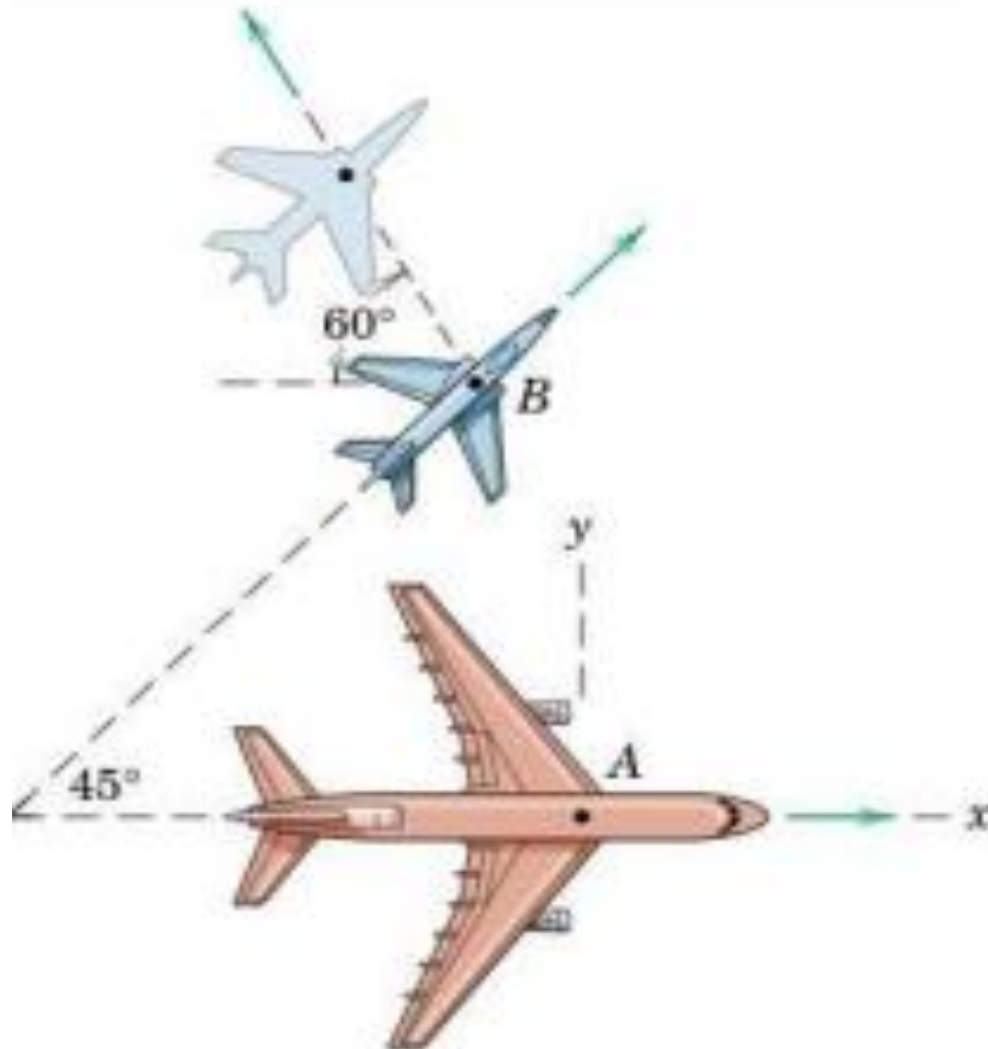
- The selection of the moving point B for attachment of the reference coordinate system is arbitrary(random).
- As shown in the figure below point A could be used as well for the attachment of the moving system, in which case the three corresponding relative motion equations for position, velocity and acceleration are

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} \quad \mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

$$\mathbf{r}_{B/A} = -\mathbf{r}_{A/B}, \mathbf{v}_{B/A} = -\mathbf{v}_{A/B}, \text{ and } \mathbf{a}_{B/A} = -\mathbf{a}_{A/B}.$$



# Example -Jet plane A and jet plane B relative motion



# Constrained motion of connected particles

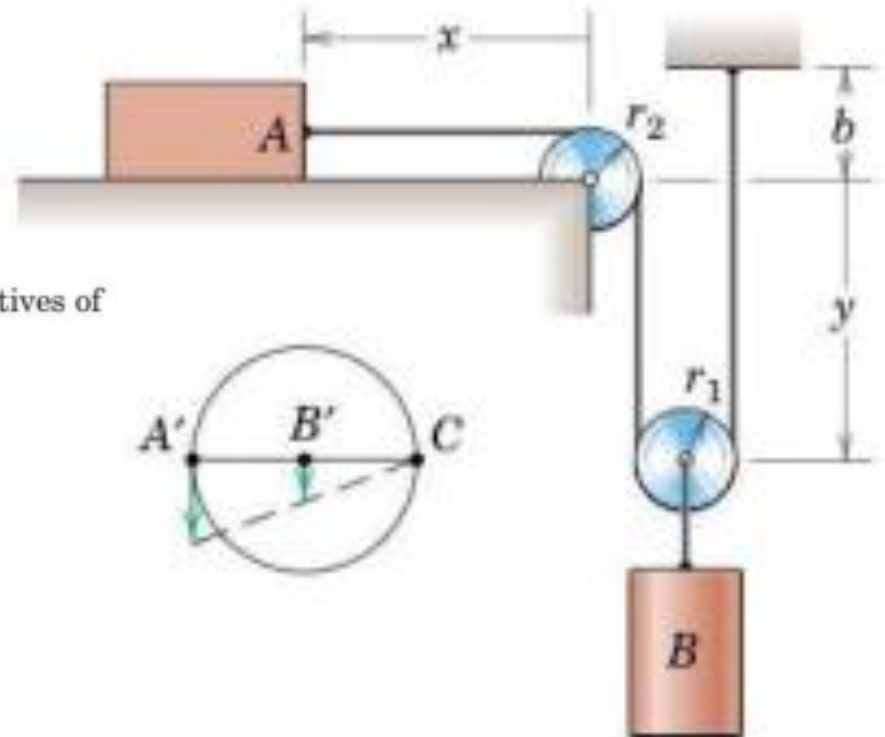
- One degree of freedom

The total length of the cable is

$$L = x + \frac{\pi r_2}{2} + 2y + \pi r_1 + b$$

With  $L$ ,  $r_2$ ,  $r_1$ , and  $b$  all constant, the first and second time derivatives of the equation give

$$\begin{array}{lll} 0 = \dot{x} + 2\dot{y} & \text{or} & 0 = v_A + 2v_B \\ 0 = \ddot{x} + 2\ddot{y} & \text{or} & 0 = a_A + 2a_B \end{array}$$



# Constrained motion of connected particles

- Two degree of freedom

The total length of the cables attached to two cylinders A and B is

$$L_A = y_A + 2y_D + \text{constant}$$

$$L_B = y_B + y_C + (y_C - y_D) + \text{constant}$$

and their time derivatives are

$$0 = \dot{y}_A + 2\dot{y}_D \quad \text{and} \quad 0 = \dot{y}_B + 2\dot{y}_C - \dot{y}_D$$

$$0 = \ddot{y}_A + 2\ddot{y}_D \quad \text{and} \quad 0 = \ddot{y}_B + 2\ddot{y}_C - \ddot{y}_D$$

Eliminating the terms in  $\dot{y}_D$  and  $\ddot{y}_D$  gives

$$\dot{y}_A + 2\dot{y}_B + 4\dot{y}_C = 0 \quad \text{or} \quad v_A + 2v_B + 4v_C = 0$$

$$\ddot{y}_A + 2\ddot{y}_B + 4\ddot{y}_C = 0 \quad \text{or} \quad a_A + 2a_B + 4a_C = 0$$

